GG-P-001278a(DSA-DM)
January 2, 1970
SUPERSEDING
GG-P-001278(DSA-DM)
December 5, 1966

INTERIM FEDERAL SPECIFICATION

PLASTIC-FILLING INSTRUMENTS AND CONTOURING INSTRUMENT, DENTAL

This Interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal Specification.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental instruments suitable for manipulating and contouring plastic-filling materials.

1.2 Classification. The plastic-filling and contouring instruments shall be of the following classes, types, and styles, as specified (see 6.1):

Class 1. Single-Ended
  Type I. Gregg
    Style 1. No. 1
    Style 2. No. 2
    Style 3. No. 3

Class 2. Double-Ended
  Type I. Ladmore
    Style 1. No. 3

FSC 6520

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Type II. Silicate (Porcelain)
Style 1. No. 1-2
Style 2. No. 5-7
Style 3. No. FP 2A

Type III. Woodson
Style 1. No. 1
Style 2. No. 2
Style 3. No. 3

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and drawings. The following specifications, standards, and drawings, of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526 Instruments, Dental and Surgical; General Specifications.
PPP-B-566 Boxes, Folding, Paperboard.
PPP-B-585 Boxes, Wood, Wirebound.
PPP-B-601 Boxes, Wood, Cleated-Plywood.
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 Box, Fiberboard.
PPP-B-676 Boxes, Setup.

STANDARDS

Federal

FED-STD-123 Marking For Domestic Shipment (Civilian Agencies).
FED-STD-151 Metals; Test Methods.

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-P-116	Methods of Preservation.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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DRAWINGS

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

20409	Pluggers, Plastic Filling, Dental, Gregg.
20650	Pluggers, Plastic Filling, Dental, Woodson.
20697	Plugger, Plastic Filling, Dental, Ladmore, No. 3.
24147	Contouring Instrument, Matrix, Dental.
20134	Pluggers, Plastic Filling, 1-2 and 5-7.

(Copies of Military specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bid or requests for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation E 18	Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM Designation E 92	Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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3. REQUIREMENTS

3.1 General requirements. The plastic-filling instruments and contouring instrument shall be in accordance with the applicable requirements of GG-I-526 and as specified herein.

3.2 Material. The instruments shall be fabricated from class 4, type 440A martensitic corrosion-resisting steel.

3.3 Style, design, and dimensions. The style, design, and dimensions of the instruments shall conform to the drawings listed in Table I for the applicable class, type, and style of instrument, and for the respective design and instrument number.

Table I. Drawing numbers

Class	Type	Style	Design	Instrument number	Drawing number
1	I	1, 2, and 3	Gregg (GRE)	1, 2, and 3	20409
2	I	1	Ladmore (LAD)	3	20697
2	II	1 and 2	Not applicable	1-2 and 5-7	20134
2	II	3	Not applicable	FP 2A	24147
2	III	1, 2,and 3	Woodson (WDS)	1, 2, and 3	20650

The instruments shall be suitable for manipulating plastic filling materials. Class 2, Type II, Style 3 instruments shall also be suitable for contouring proximal surfaces of dental amalgam matrix bands.

3.4 Handle serrations or knurls. Instrument handles shall have 40 ± 10 fillet-type serrations or notch-type serrations per inch. The depth of the serrations shall be 0.002 to 0.004 inch. The alternate handles of Class 2, Type II, Styles 1 and 2 instruments shall have 30-degree or 45-degree diamond knurls.

3.5 Hardness. The operative ends and shanks of the instruments shall have a Rockwell hardness of C52 to C56 after proper stress relieving, and the handles shall have a Rockwell hardness of C20 minimum, when tested as specified in 4.4.2.

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3.6 Corrosion resistance. The instruments shall be properly passivated, and shall pass the boil test and the copper sulfate test requirements specified in GG-I-526.

3.7 Finish.

3.7.1 Operative ends and shanks. Operative ends and shanks of the instruments shall have a mirror or bright polished finish with smooth tips, except that the front face of blade-type operative ends on Class 2, Type III instruments shall have a fine honed finish, and the sides of cone-type operative ends on Class 2, Type III instruments shall have a mirror or fine honed finish. The fine honed finish specified shall be a finish such as is produced by using grade FF emery on a leather wheel.

3.7.2 Handles. The handles of the instruments shall have a bright polished finish. The alternate handles for Class 2, Type II, Styles 1 and 2 instruments shall have a fine knurled or ground finish.

3.8 Identification marking. The handles of the instruments shall be permanently and legibly marked as specified in GG-I-526, and shall also be marked with the applicable instrument design (or abbreviation thereof) and instrument number as shown in Table I. For single-ended instruments (Class 1) the respective instrument number shall be placed no less than 5 millimeters (mm.) nor more than 12 mm. from the proximal end (free end). For double-ended instruments (except Class 2, Type II, Styles 1 and 2) the respective instrument number shall be placed to the right of the name of the design of the instrument. For Class 2, Type II, Styles 1 and 2 instruments having an octagonal handle, the respective instrument numbers shall be placed on the handle no less than 5 mm. nor more than 12 mm. from the junction of the handle and shank of the applicable operative end. For the alternate handle design on Class 2, Type II, Styles 1 and 2 instruments, the respective instrument numbers (1-2 or 5-7) shall be placed near the center of the handle and positioned toward the applicable operative end.

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3.9 Workmanship. Workmanship shall be first class throughout. The instruments shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the name of the manufacturer or supplier, the name of the material, the lot number, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables II and IV (see 4.2.1 and 4.3) and Table III (see 4.2.2).

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. The unit of product for examination purposes shall be one instrument.

Table II. Inspection and Acceptable Quality Levels

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one instrument.

Table III. Test Criteria

Characteristic	Requirement	Test procedure	Inspection level
Chemical analysis	3.2	4.4.1	(see 4.2.3)
Hardness	3.5	4.4.2	S-2
Boil test	3.6 and GG-I-526	GG-I-526	S-2
Copper sulfate test	3.6 and GG-I-526	GG-I-526	S-2

4.2.3 Sampling of raw materials. A sufficient quantity of each lot of raw steel used in the fabrication of the instruments shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination. The dental instruments shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in Table IV.

Table IV. Classification of defects

Category	Defect
Major A:	
101	Class or type of instrument not as specified.
102	Style and design of instrument not as shown on applicable drawing.
103	Working surfaces of instrument not free from nicks, feathers, pits, burrs, or corrosion.
104	Finish on operative end(s) not as specified.
105	Instrument not free from cracks.
Major B:	
151	Dimensions of operative end(s) not as shown on applicable drawing.
Minor A:	
201	Handle serrations (or knurling, when applicable) of incorrect type, or missing altogether.
202	Finish on shank(s) and handle not as specified.
203	Instrument not free from foreign matter, dirt, oil, or polishing compound.
204	Instrument not free from coarse grind marks or scratched surface areas.
205	Identification marking not present or not permanent, legible, and complete.
206	Packaging and packing not as specified.
207	Marking on packaging and pack not as specified

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Table IV. Classification of defects (cont'd.)

Category	Defect
Minor B:	
251	Any dimension (other than those in 151) not as specified in applicable drawing.
252	Number or depth of serrations not within range specified.
253	Instrument number(s) improperly located.

4.4 Test procedures.

4.4.1 Chemical analysis. In the absence of certificates of quality, the chemical analysis of the instruments shall be determined in accordance with method 111 or 112 of FED-STD-151. In the event of conflict, method 111 shall govern.

4.4.2 Hardness. The hardness shall be determined in accordance with ASTM Designation E18 or E92.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, or C, as specified (see 6.1).

5.1.1 Unit of issue. One plugger, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units will be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. No preservative is required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IA-8.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable; or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers, of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and shall be constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers, of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

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5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Class, type, and style required (see 1.2).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, styles, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

6.3 Specification changes. The margins of this specification have not been annotated with asterisks to indicate where changes from the previous issue have been made due to the extensive changes to this document.

Custodian:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION: GG-P-001278a(DSA-DM) FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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